Dataset Used: birthwt (2).csv

```
Regression Type: LOGISTIC
Response Variable: low
Predictors: ['smoke', 'ht', 'ui']

=== LOGISTIC REGRESSION RESULTS ===
                           Logit Regression Results
==============================================================================
Dep. Variable:                    low   No. Observations:                  189
Model:                          Logit   Df Residuals:                      185
Method:                           MLE   Df Model:                            3
Date:                Sun, 25 Oct 2026   Pseudo R-squ.:                 0.06250
Time:                        23:27:19   Log-Likelihood:                -110.00
converged:                       True   LL-Null:                       -117.34
Covariance Type:            nonrobust   LLR p-value:                  0.002125
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -1.3551      0.241     -5.611      0.000      -1.828      -0.882
smoke          0.6831      0.329      2.075      0.038       0.038       1.329
ht             1.4174      0.624      2.273      0.023       0.195       2.640
ui             1.0384      0.428      2.426      0.015       0.200       1.877
==============================================================================

Pseudo R-squared (McFadden): 0.0625

Confusion Matrix:
[[119  11]
[ 45  14]]
Accuracy: 0.7037

Classification Report:
              precision    recall  f1-score   support

           0     0.7256    0.9154    0.8095       130
           1     0.5600    0.2373    0.3333        59

    accuracy                         0.7037       189
   macro avg     0.6428    0.5763    0.5714       189
weighted avg     0.6739    0.7037    0.6609       189
```

Dataset Used: places_2025 (2).csv

```
Regression Type: POISSON
Response Variable: mhlth_crudeprev
Predictors: ['binge_crudeprev', 'checkup_crudeprev', 'csmoking_crudeprev',
'depression_crudeprev', 'obesity_crudeprev', 'sleep_crudeprev',
'selfcare_crudeprev', 'indeplive_crudeprev', 'loneliness_crudeprev',
'foodinsecu_crudeprev', 'housinsecu_crudeprev']
```

```
=== POISSON REGRESSION RESULTS ===
               Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:        mhlth_crudeprev   No. Observations:                 2299
Model:                            GLM   Df Residuals:                     2287
Model Family:                 Poisson   Df Model:                           11
Link Function:                    Log   Scale:                          1.0000
Method:                          IRLS   Log-Likelihood:                -5397.6
Date:                Sun, 25 Oct 2026   Deviance:                       39.316
Time:                        23:29:37   Pearson chi2:                     39.2
No. Iterations:                     4   Pseudo R-squ. (CS):             0.1925
Covariance Type:            nonrobust
==============================================================================
=========
                           coef    std err          z      P>|z|      [0.025
0.975]
------------------------------------------------------------------------------
---------
const                    1.7032      0.172      9.888      0.000       1.366
2.041
binge_crudeprev          0.0032      0.003      0.954      0.340      -0.003
0.010
checkup_crudeprev       -0.0004      0.002     -0.192      0.848      -0.004
0.004
csmoking_crudeprev       0.0043      0.004      1.100      0.271      -0.003
0.012
depression_crudeprev     0.0137      0.002      5.800      0.000       0.009
0.018
obesity_crudeprev        0.0001      0.002      0.076      0.940      -0.003
0.004
sleep_crudeprev          0.0024      0.002      1.059      0.289      -0.002
0.007
selfcare_crudeprev      -0.0608      0.022     -2.708      0.007      -0.105
-0.017
indeplive_crudeprev      0.0522      0.014      3.759      0.000       0.025
0.079
loneliness_crudeprev     0.0113      0.003      3.717      0.000       0.005
0.017
foodinsecu_crudeprev    -0.0083      0.007     -1.247      0.212      -0.021
0.005
housinsecu_crudeprev     0.0150      0.008      1.786      0.074      -0.001
0.032
==============================================================================
=========

Pseudo R-squared (Deviance-based): 0.9260

--- Variance Inflation Factor (VIF) ---
               Feature        VIF
```

```
0         binge_crudeprev   2.528842
1      checkup_crudeprev   1.750444
2     csmoking_crudeprev   6.934794
3   depression_crudeprev   2.216376
4      obesity_crudeprev   2.573017
5        sleep_crudeprev   3.209334
6     selfcare_crudeprev  30.514413
7    indeplive_crudeprev  30.953309
8   loneliness_crudeprev   3.432565
9   foodinsecu_crudeprev  62.830787
10  housinsecu_crudeprev  46.839954
11                 const   1.000000
```

# For the Logistic Regression (birthwt (2).csv)

1. What is the model's accuracy?
    - 70.37% (or 0.7037)
2. What is the model's lowest precision?
    - 0.5600 (associated with class 1; class 0 has a precision of 0.7256)
3. Do all categories have at least 30 support? (confusion matrix)
    - Yes. Category 0 has a support of 130 and Category 1 has a support of 59. Both exceed 30 support.
4. Based on the pseudo-$R^2$ value, is this model good in this scenario?
    - No. The McFadden Pseudo-$R^2$ is 0.0625 (6.25%). This is a very low value, indicating that the model explains very little of the variance in low birthweight and provides a weak fit.

# For the Count-Variable Regression (places_2025 (2).csv)

1. Your model should include 4 significant predictors. What are they?
    - At the $\alpha = 0.05$ significance level ($P > \vert{}z\vert{} < 0.05$), the 4 significant predictors are:
        - depression_crudeprev ($P = 0.000$)
        - selfcare_crudeprev ($P = 0.007$)
        - indeplive_crudeprev ($P = 0.000$)
        - loneliness_crudeprev ($P = 0.000$)
2. Which of these predictors is the most significant (hint: look at the 'coef' column)?
    - selfcare_crudeprev has the largest absolute coefficient magnitude ( $coef = -0.0608$) among the significant predictors, representing the largest proportional change in the response per unit increase.
3. Do any of your significant predictors have an overinflated VIF (VIF > 10)?
    - Yes. Two of the significant predictors exceed the threshold of 10:
        - selfcare_crudeprev ($VIF = 30.51$)
        - indeplive_crudeprev ($VIF = 30.95$)

4. How much of the variability is captured by the model (hint: look at pseudo-$R^2$)?
    - 92.60% (Deviance-based Pseudo-$R^2$ of 0.9260), indicating that a very high proportion of the variability is captured by the model.